

Course: Networked Control

Lecture III.1: Introduction to networked control systems

Prof. Marcello Farina

Dipartimento di Elettronica, Informazione e Bioingegneria (DEIB)

Politecnico di Milano, Italy

`marcello.farina@polimi.it`

Networked control: from Part II to Part III

What is this course about?

Issues arising in the **design and implementation** of
control systems
characterized by the presence of
components arranged in a networked
configuration

Networked control: from Part II to Part III

Issues

MIMO systems

Very complex and large-scale system under control

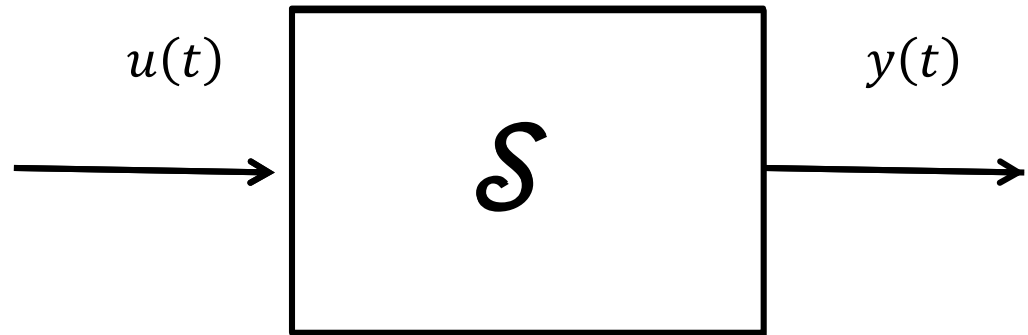
- Large number of interacting subsystems
- Complex system structure



High-dimensional and complex control problem.



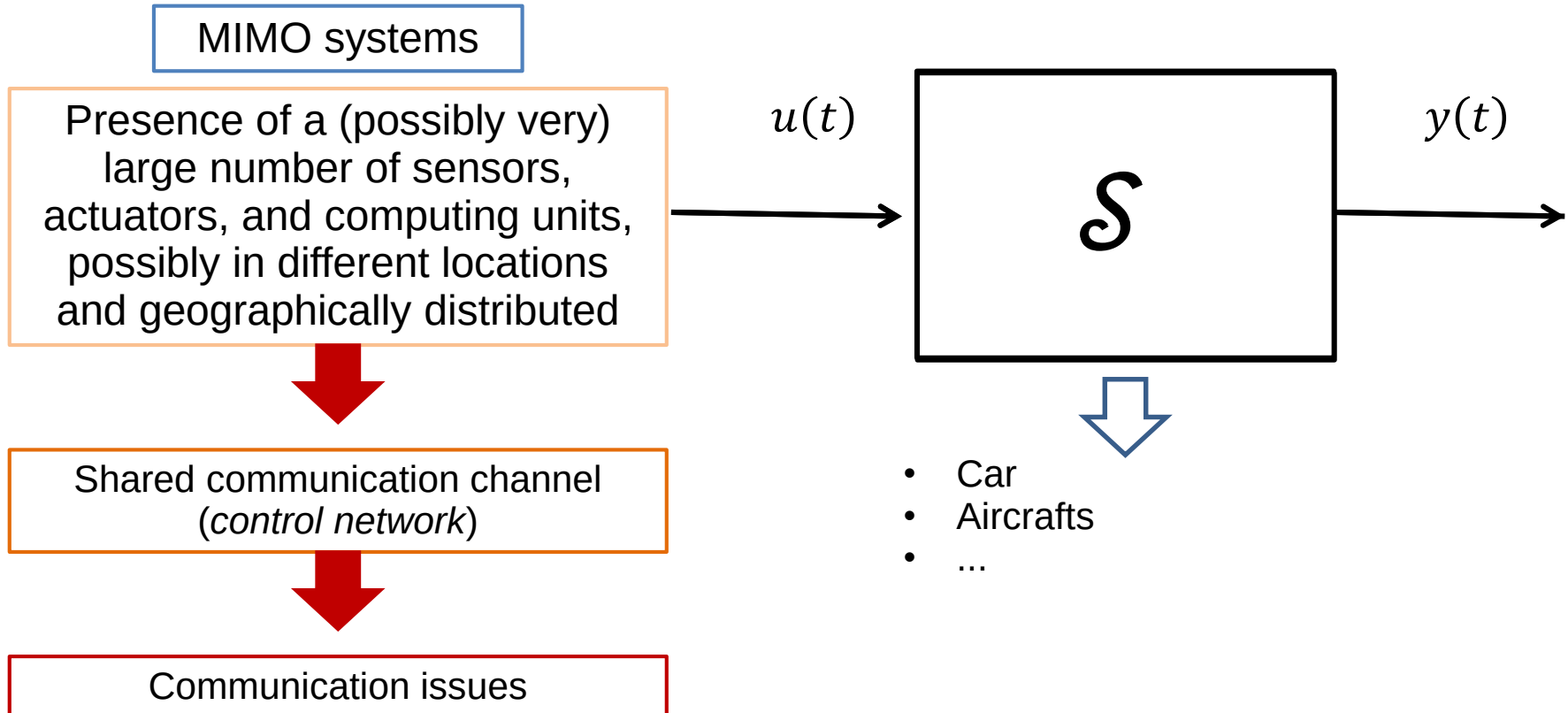
- Need of «non centralized» control schemes
 - Subsystem/device level
 - Coordination between subsystems.



- Chemical plant
- Electric grid
- Fleet of autonomous underwater vehicles
- Irrigation network

Networked control: from Part II to Part III

Issues

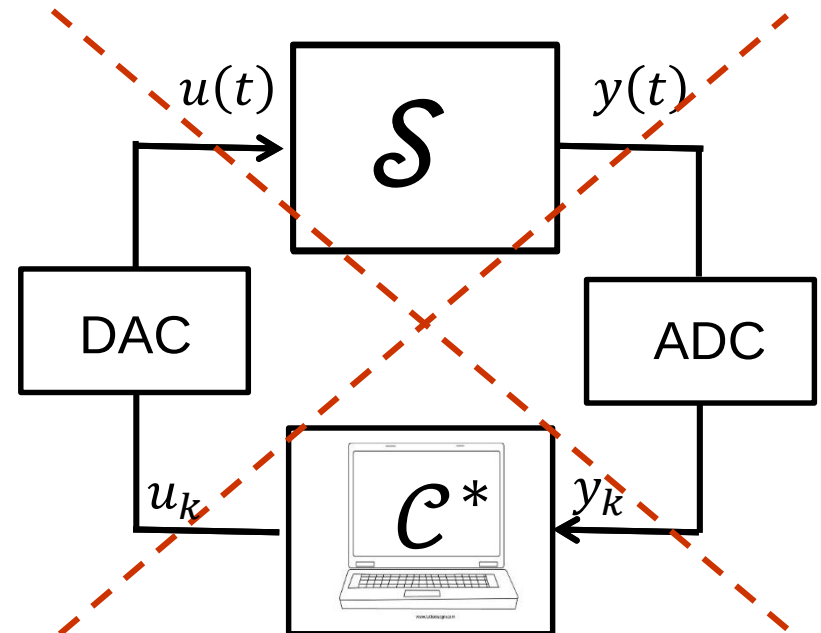


Networked control: from Part II to Part III

Issues

New control applications pose new **challenges**.

Classic **centralized** control solutions with **ideal communication** may be not feasible/possible/adequate



Networked control: from Part II to Part III

What is this course about?

Issues arising in the **design and implementation** of
control systems
characterized by the presence of
**components arranged in a networked
configuration**



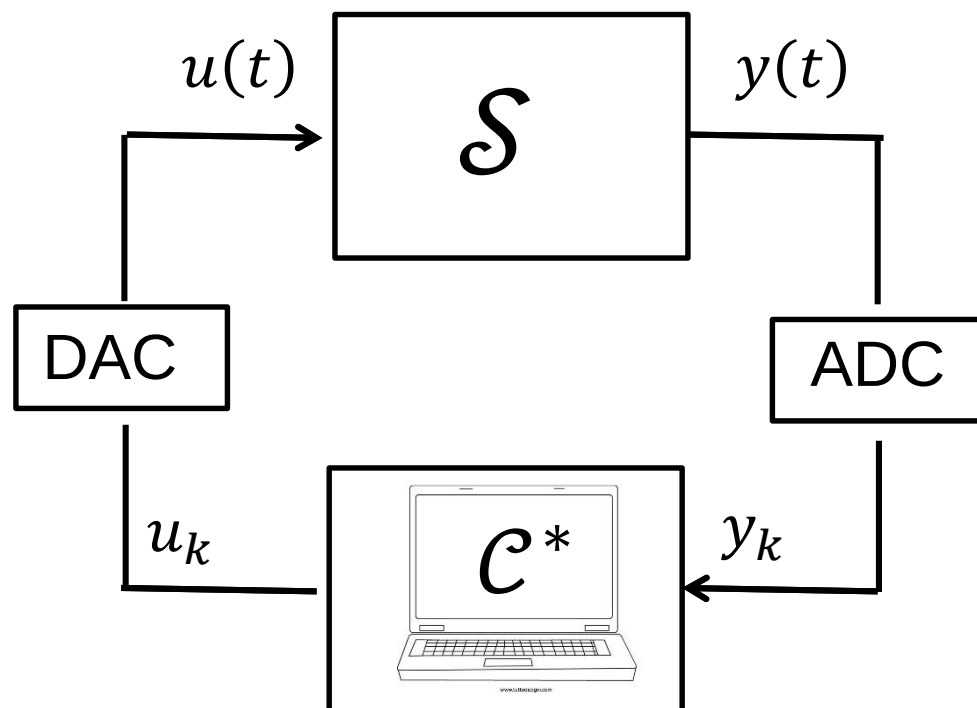
*Decentralized and
distributed control
(Part II).*



*Control under
non-ideal
communication
(Part III)*

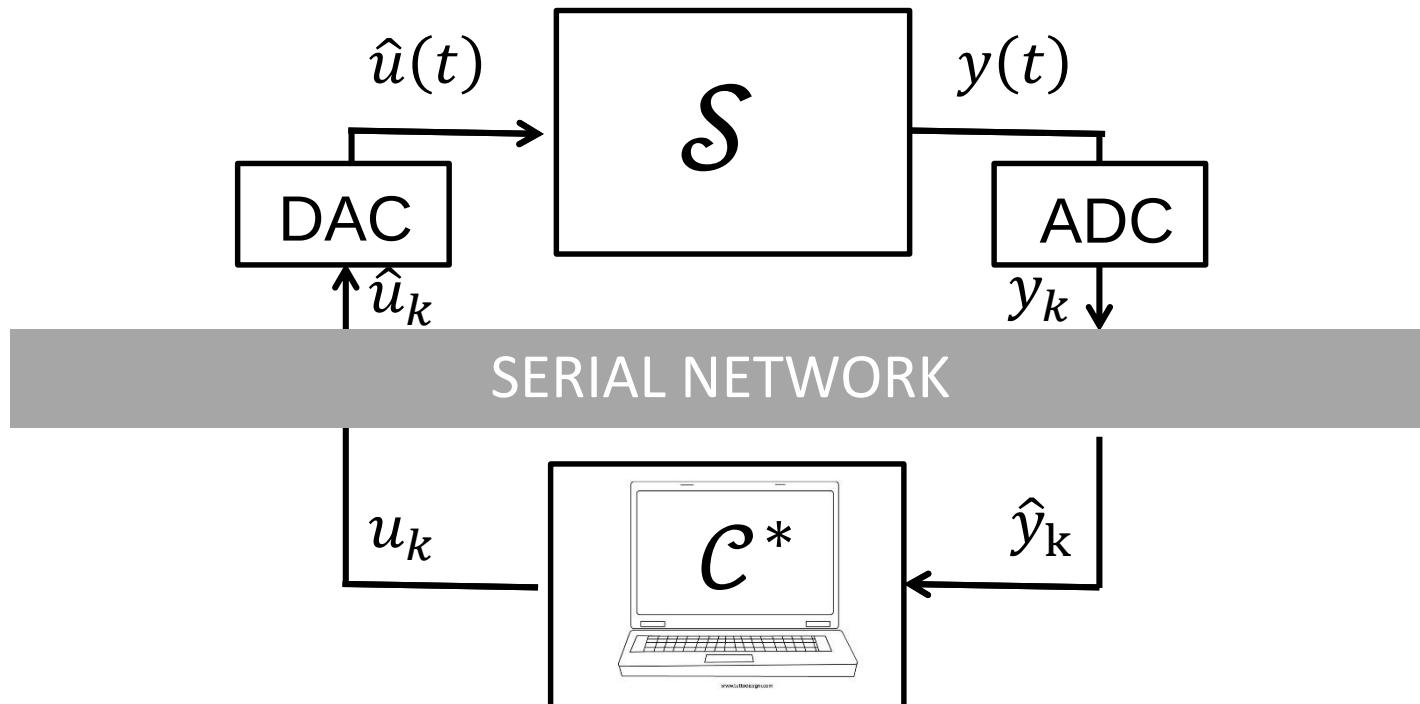
Networked control systems

- In the next part of this course we will deal with **control over shared serial communication channels**.

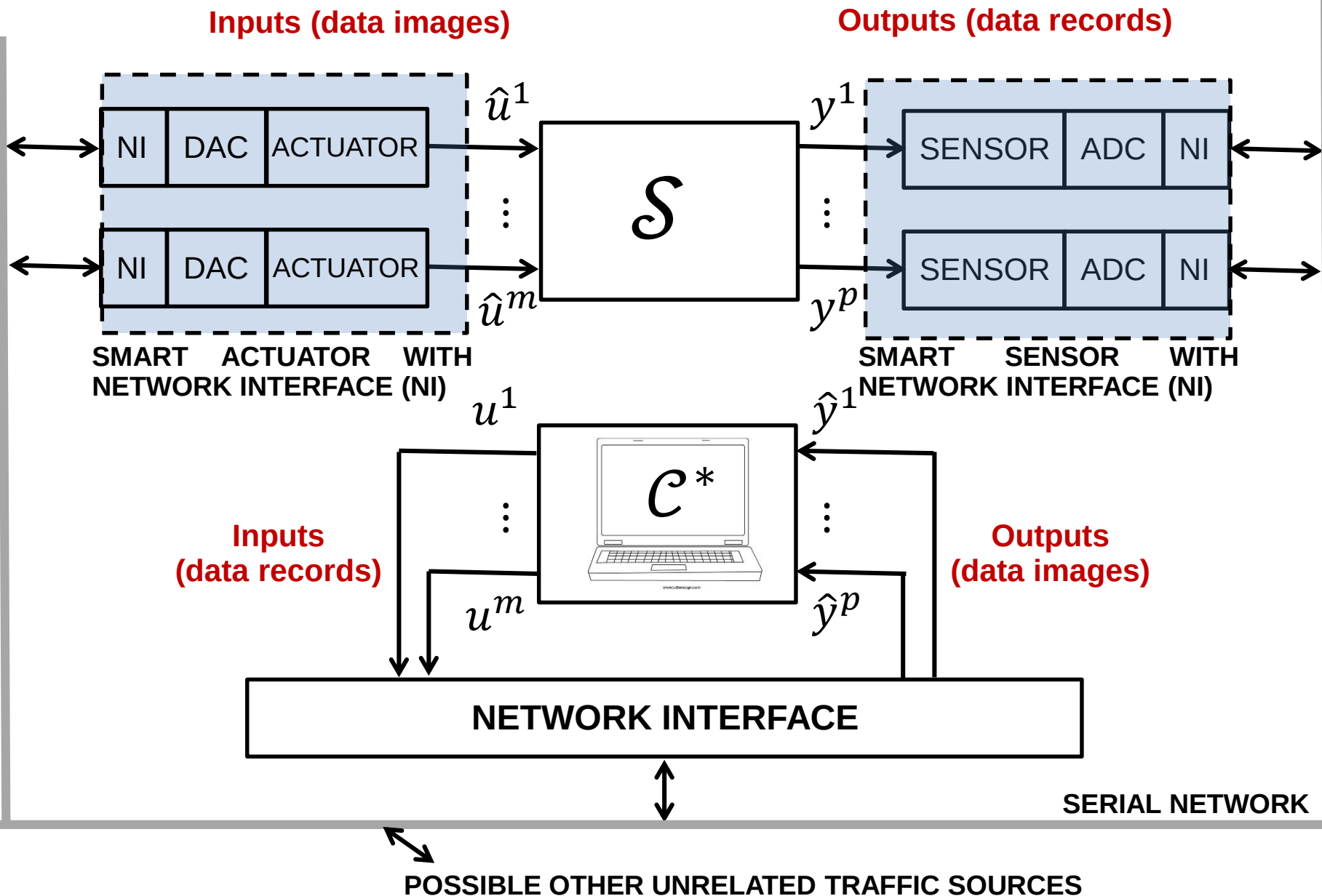


Networked control systems

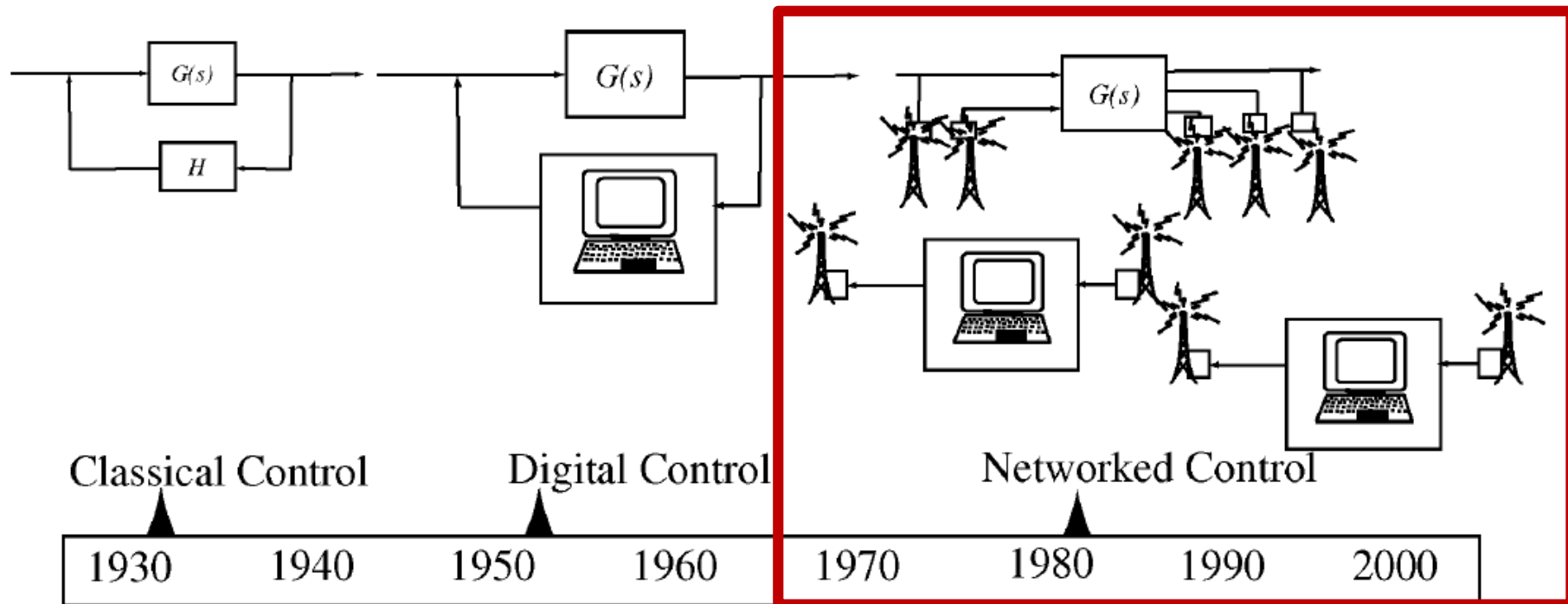
- In the next part of this course we will deal with **control over shared serial communication channels**.
- We define as a **networked control system** a **control system** whose parts (sensors, actuators, controllers) are connected through a serial communication network.
- A network is used, e.g., for communicating data
 - from the sensors to the controller, and/or
 - from the controller to the actuator



Networked control systems



Historical background



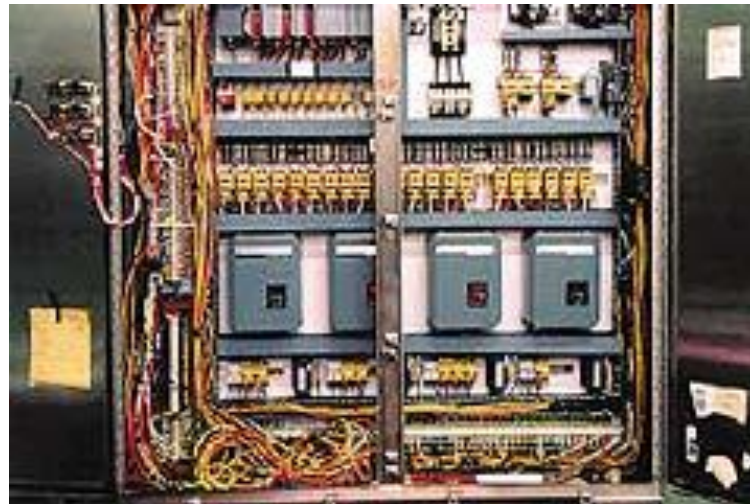
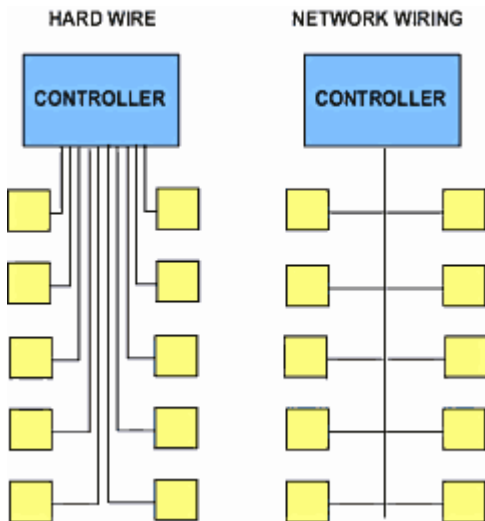
From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

Historical background

- The beginning of research on **network control systems** can be traced back to 1983.
- In 1983 Bosch GmbH launched a feasibility study for the use of devices, connected using an *ad hoc* network, to control different functions in passenger cars. This was the earliest effort towards modern networked control.
- In 1986 the Control Area Network (CAN) protocol was announced at the Congress of the Society of Automotive Engineers, Detroit.
- In 1987 the CAN hardware (in the form of INTEL 82526 chip) was introduced.
- Today all cars manufactured in Europe embed systems connected through CAN.
- Networked control system technologies (with different protocols and network types) are nowadays fairly mature in a variety of industrial applications, e.g.,
 - Automobiles
 - Manufacturing plants/industrial automation
 - Aircrafts/spacecrafts
 - Heating, ventilation, and air conditioning (HVAC) systems
- Recent trends: integration through wireless rather than wired communication channels.

Benefits of networked control systems

- Often it is the **only technical solution**, e.g., when distances are large.
- **Cost**: save wiring when multiple devices are connected together.
 - A serial network consists of a single trunk line instead of hundreds of individual wires.
 - As the system grows past the deadline of 100 I/O points the added expense of network hardware is offset by savings in wiring time. This benefit grows exponentially as the size of the system.



Control panel with

- Point-to-point wired communication
- Devicenet

Courtesy from <https://www.perrymarshall.com/articles/industrial/part-1/>

Benefits of networked control systems

- **Modularity:** connectivity is achieved through software. It does not matter in which place of the network a component is connected.
 - Modularization of functionalities: decision and control functions can be distributed among different controllers in the network.
 - Availability of standard interfaces for interoperability and interchangeability.
 - Reconfigurability (simple installation and maintenance).
 - Plug and play functionalities.
 - Self configuration: controllers can detect which devices are connected and load suitable software configurations.
- **Diagnostics capabilities:** information (e.g., about malfunctionings of single devices) can be transmitted by the sensors and are made accessible through the network.
- Avoid single point-of-failure and **allow for redundant configurations.**

Challenges of networked control systems

(for wired, but especially wireless communication networks)

- **Band-limited channels:** constraint on the amount of information (number of packets and/or number of bits per packet) that can be transmitted per time unit.
 - Particularly critical in
 - Power-starved vehicles (planetary rovers).
 - Long endurance, energy-limited systems (sensor networks).
 - Underwater vehicles, micro-sensors, and micro-actuators.
 - This has a major impact on the **tradeoff between sampling and quantization**.
- **Delays** (possibly time-varying), due to
 - Congestions.
 - Competition for communication resources, that depends on the network protocol.
 - Transmission time, i.e., the time that a packet spends in the network.

Challenges of networked control systems

(for wired, but especially wireless communication networks)

- **Packet dropouts:** loss of packets during the transmission. Due to:
 - Errors in nodes and physical network links (more common in wireless than in wired networks).
 - Buffer overflow during congestions.
 - Long transmission delays or scheduling time. The receiver may discard the old data in real-time control applications.
- Noise.
- Fading, i.e., variation of the attenuation of a signal (with respect to, e.g., geographical position and radio frequency).

Challenges of networked control systems

Disclaimer:

In this course we will deal with

issues arising in control over network

e.g., impact of bandwidth limitations, delays, and packet dropouts on the stability properties of the networked control system.

We are **not interested in**

control of communication networks

e.g., maximization of the communication network throughput, development of efficient scheduling/routing algorithms.

Main related references

- G.C. Walsh, H. Ye, Scheduling of networked control systems. *IEEE Control System Magazine*, 21(1), 2001, pp. 57-65.
- R. J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28.
- Perry Sink Marshall, A comprehensive guide to industrial networks. <https://www.perrymarshall.com/articles/industrial/part-1>